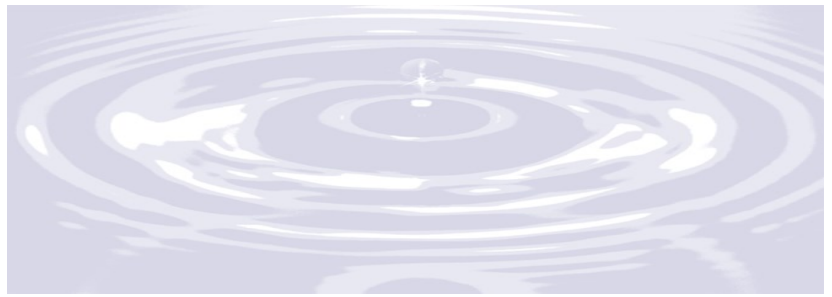


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Summary Letter Site Plan Review and Notice of Intent EMD Millipore Corporation 12 and 17 Cherry Hill Drive *Danver, Massachusetts 01923* *Map 37 Lot 38 and Map 29 Lot 1*

PREPARED FOR:

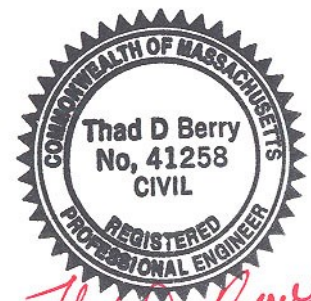
EMD Millipore Corporation

12 and 17 Cheery Hill Drive
Danvers, MA. 01923

PREPARED BY:

ASB design group, llc

363 Boston Street, Route 1
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Job No. 2022-30



Thad D Berry

June 2024

ASB design group

Town of Danvers Planning Board and Conservation Commission
Town of Danvers Town Hall
1 Sylvan Street
Danvers, MA. 01923

Date: June 6, 2025

Re: **Summary Letter – Site Plan Review and Notice of Intent
12 and 17 Cherry Hill Drive
EMD Millipore Corporation Danvers, MA. 01923
Job No. 2022-30
Map 37 Lot 38 and Map 29 Lot 1
MADEP No.: TBD Notice of Intent**

Dear Members:

On behalf of our client EMD Millipore Corporation, **ASB design group, LLC (ASB)** is submitting this Summary Letter for the Site Plan Review and Notice of Intent regarding the redevelopment of 12 and 17 Cherry Hill Drive for your review, comment, and approval.

The applicant will be preparing submittals for the following approvals:

- Site Plan Review – Town of Danvers Planning Board
- Notice of Intent – Town of Danvers Conservation Commission and The Massachusetts Department of Environmental Protection (MassDEP)
- Stormwater Management Permit – Town of Danvers Department of Public Works
- Federal – NPDES General Site Permit (greater than 1 acre of total land disturbance).

This submittal includes:

- Summary Letter
- Operation and Maintenance Plan for Construction Phase
- Operation Maintenance Plan for Post Construction Phase

Appendix A

- Figure 1: Existing Conditions Aerial Map – Overall Site (12 and 17 Cherry Hill Drive)
- Figure 1A: Existing Conditions Aerial Map – Project Site
- Figure 2: Proposed Conditions Aerial Map
- Figure 3: Soils and Hydrologic Soils Map
- Figure 4: 2008 – 2009 Aerial Image
- Figure DP1 Pre- Construction Hydrology
- Figure DP2 Post Construction Hydrology

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- TSS Work Sheet
- HydroCAD Print Out ¹
- Massachusetts Department of Environmental Protection Stormwater Checklist Summary
- Massachusetts Department of Environmental Protection Stormwater Checklist

Accompanying the Summary Letter

- Sheets C1 – C14 (see Sheet C1 for Drawing List: Site Plan Review and Notice Intent Plans)

1 [HydroCAD Node Listing and Totals are provided for the 2, 10, 25, 50 and 100-year storm events.](#)

Design Team

The Design Team for the proposed project includes:

- Attorney
Mann and Mann, P.C.
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- Robert Autenzio
Operations Manager
Paul's Landscape Service & Supplies LTD
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978-649-9932

PROJECT OVERVIEW

Existing Conditions

The project sites are located on Cherry Hill Drive. 17 Cherry Hill Drive is located on the easterly side of the roadway (Lot Area = 8.781 acres) and 12 Cherry Hill Drive on the westerly side of the roadway (Lot Area = 5.046 acres) as shown in Figure 1. The manufacturing facility and accessory parking is located at 17 Cherry Hill Drive. 12 Cherry Hill Drive is used solely for parking that is an accessory to the manufacturing use at 17 Cherry Hill. Please note that all handicapped parking spaces are located at 17 Cherry Hill Drive at the entrance to the building. Figure 1A shows the project re-development area. Currently Millipore leases offsite parking spaces at 54 Cherry Hill, which lease is expiring at the end of 2025.

Wetland Resource Areas

Norse Environmental Services, Inc. flagged the bordering vegetated wetlands and top of riverbank in 2024-25. The state and local wetland resource area buffer zones are shown on Sheets C2 to C5. They are:

- 35' No Disturbance Zone (Town of Danvers Setback by Law)
- 50' No Build Zone (Town of Danvers Setback by Law)
- 100' Buffer Zone to BVW (State)
- 100' Inner Riparian (State)
- 200' Riverfront Area (State)

The upland area that will be developed does not lie within a flood hazard area or aquifer protection district.

Soils

The Natural Resources Conservation Service (NRCS) Soils and Hydrologic Soils Group Map depicts the soil classification for the onsite soils (see Figure 3). There are 3 soil types that fall with the project site. They are:

- 43A: Scarboro Mucky Fine Sandy Loam (0 to 3 percent slopes) with a Hydrologic Soil Group (HSG) Classification of A/D.
- 602: Urban Land - Hydrologic Soil Group (HSG) Classification of N/A.

- 651: Udorthents, Smoothed - Hydrologic Soil Group Classification of N/A.

Soil testing was conducted at the site on November 15, 2024, May 2, 2025, and May 15, 2025. The purpose of the soil testing was to determine the suitability for the design of stormwater management Best Management Practice Ponds including stormwater infiltration. The soil testing locations are shown on Sheets C2 – C5. A total of 8 deep hole tests were excavated on site and 5 soil cores. The soil logs and soil cores are shown on Sheet C3 along with the NRCS Soil Classifications.

Proposed Conditions

As further described below, Millipore is proposing to add one hundred forty-six (146) parking spaces at 12 and 17 Cherry Hill Drive as shown in Figure #2 and Sheets C3 – C5. Currently, the number of available parking spaces is not sufficient to support Millipore's existing workforce, which is particularly problematic during shift changes, when employees are forced to park on the grass areas at 12 Cherry Hill Drive (see Figure 1A and Sheet C2). The additional parking is intended to: (i) address the current parking shortage; (ii) replace the offsite parking at 54 Cherry Hill; and (iii) create new parking spaces to accommodate Millipore's growing manufacturing business and expanding workforce. Millipore is proposing to create three (3) new parking areas (Parking Areas A, B and C) that will support the additional spaces. Each parking area is summarized below.

Parking Area A

Parking Area A is located at 17 Cherry Hill Drive as shown on Sheet C4. A total of fifty-five (55) parking spaces are being proposed in this location. This parking area abuts the manufacturing facility. A 24-foot-wide access to this parking area extends off the existing parking lot that parallels the manufacturing facility. Each parking space will be 8.5' wide by 18' long with 24' travel aisles.

Parking Area A falls within the 35' No Disturb Zone, 50' No Build Zone, which areas are within 100' Buffer Zone and 200' River Front Area as shown on Sheet C4. Currently these areas are developed with lawn, gravel walkway and an existing Stormwater BMP. Figure #4 (2008 – 2009) shows that this area was disturbed with lawn and the stormwater BMP.

The work within the 35' No Disturb Zone and 50' No Build Zone will be for a Constructed Stormwater Wetland BMP. Approximately 8 square feet of pavement will fall within the 50' No Build Zone (See Figure #2). The Constructed Stormwater Wetland BMP A (see Sheet C6) will provide for a Total Suspended Soil Removal (TSS) of 85% (see TSS Removal Work Sheet).

In addition, the existing Stormwater Water BMP will be improved as outlined below.

Parking Area B

Parking Area B is located at 12 Cherry Hill Drive as shown on Sheet C5. A total of twenty-seven (27) parking spaces are being proposed in this location. Parking Area B is located within the grassed area where cars are currently forced to park during shift changes and where BMP B is located (see Figure 1A and Sheet C2).

A 24-foot-wide access aisle to this parking area extends off the existing entrance to the parking lot at 12 Cherry Hill Drive. Each parking space will be 8.5' wide by 18' long with 24' travel aisles.

The work for Parking Area B falls within the 35' No Disturb Zone, 50' No Build Zone, 100' Buffer Zone and 100' Inner Riparian Zone shown on Sheet C5. Currently these areas are developed with lawn and pavement. Figure #4 (2008 – 2009) shows that this area was disturbed with lawn.

The entire proposed paved parking area for Parking Area B is outside of the 50' No Build Zone and 100' Inner Riparian Zone. Only the work for the Constructed Stormwater Wetland BMP B (see Sheet C7) falls within the 35' No Disturb Zone, 50' No Build Zone and 100' Inner Riparian Zone shown on Sheet C5. The Constructed Stormwater Wetland BMP B will provide for a Total Suspended Soil Removal (TSS) of 85% (see TSS Removal Work Sheet).

Parking Area C

Parking Area C is located at 12 Cherry Hill Drive as shown on Sheet C5. A total of fifty-three (53) parking spaces are being proposed in this location. This parking area is located at the rear of the 12 Cherry Hill Drive parking lot. In addition, there will also be a total of eleven (11) parallel parking spaces being proposed along 24' wide access aisle for the 12 Cherry Hill Drive parking lot. Currently cars parallel park in this location. This area is comprised of a dense grade surface. Winter plowing displaces the dense graded material into the existing stormwater BMP that parallels this area.

In the spring of 2025, Paul's Landscaping Services cleaned the existing stormwater BMP of gravel/dense graded material. At that time invasive species (Phragmites) were removed and the BMP was stabilized with loam and seed. Please note that the Phragmites have reemerged and will need further removal.

A 24-foot-wide access aisle to Parking Area C extends off the existing parking lot. Each parking space will be 8.5' wide by 18' long with 24' travel aisles. The parallel parking spaces will be 22' long by 8' wide.

The work for Parking Area C and the parallel parking spaces falls within the 35' No Disturb Zone, 50' No Build Zone, 100' Buffer Zone, 100' Inner Riparian Zone and 200' River Front Area as shown on Sheet C5. Currently these areas are developed with lawn, a gravel walkway and an existing Stormwater BMP. Figure #4 (2008 – 2009) shows that these areas were disturbed

with lawn and the stormwater BMP's.

The entire proposed paved parking area for Parking Area C is outside of the 50' No Build Zone and 100' Inner Riparian Zone. Only the work for the Construction of Stormwater Wetland BMP C and the proposed improvements to the exiting Stormwater BMP (existing drainage swale – eastern and western sections) as outlined below, fall within the 35' No Disturb Zone and 50' No Build Zone, 100' Inner Riparian Zone and 200' River Front Area will be . The Constructed Stormwater Wetland BMP C (see Sheet C7) will provide for a Total Suspended Soil Removal (TSS) of 85% (see TSS Removal Work Sheet).

Proposed Resource Area Buffer Zone Impact Mitigation

The proposed project is a redevelopment of existing disturbed land within the 35' No Disturb Zone, 50' No Build Zone, 100' Buffer Zone, 100' Inner Riparian Zone and 200' River Front Area. As evidenced by the photographs submitted herewith, the project areas are all impacted areas. The areas have been disturbed and are maintained as lawn and stormwater BMP's. To mitigate any adverse impacts resulting from the continued use of the areas, Millipore proposes the following measures:

Invasive Species Removal

At each location (Parking Area A, B, and C) there are invasive species as noted in the Notice of Intent prepared by Norse Environmental Services, Inc. There are also invasive species within the limits of the existing stormwater BMPs at 12 and 17 Cherry Hill Drive. The invasive species encountered at the site are:

- Parking lot B & C invasive species within the swale: common reed (*Phragmites australis*)
- Invasive species along the southerly berm include multiflora rose (*Rosa multiflora*), oriental bittersweet (*Celastrus orbiculatus*), Tartarian honeysuckle (*Lonicera tatarica*) and glossy buckthorn (*Rhamnus frangula*)

As part of this redevelopment, the invasive species will be removed and these areas will be re-vegetated with native, trees, shrubs and ground covers. A long-term Operation and Maintenance Plan will include monitoring for and removal of any invasive species that may reemerge in the future.

Existing Stormwater BMP Improvement – 12 Cherry Hill Drive (see Sheet C7)

Currently there is an existing stormwater BMP that parallels the 24' wide access aisle to the parking lot. The base of the stormwater BMP averages approximately 3.5' wide with side slopes of 2.75 to 1. The top width of the BMP averages about 21.0' and has a total length is 310'. Within the stormwater BMP there were two stone riprap weirs. There is a single 18" RCP outlet

located at the rear section of the BMP. A rip-rap weir overflow (approximated elevation 54.75) is located approximately in the center of the BMP. Two (2) catch basins within the paved parking lot direct the stormwater runoff to the stormwater BMP. Due to clogged stone rip-rap weirs placed within the BMP the stormwater does not flow/drain properly. This is exacerbated by the fact that the slope of the BMP's inlets to the outlet is basically level. In addition, winter plowing has deposited silt/soil into the stormwater BMP. Currently there is no way to access the southerly side of the stormwater BMP for long term maintenance. Sheet C2 documents the conditions outlined above.

As part of the drainage and stormwater design the existing stormwater BMP will be increased in size as outlined below:

- The existing stormwater BMP was divided into the eastern and western sections based on the location of the existing rip-rap overflow. Under existing conditions, the total volume for the eastern section is 5,124 cubic feet. For the western section the total volume is 5,308 cubic feet.

Under the proposed redevelopment the eastern section's volume will be increased to 7,924 cubic feet (2,800 cubic feet increase). The western section will be increased in volume to 8,106 cubic feet (2,798 cubic feet increase).

- The base width of the eastern section will be widened to 4.0' and the western section base width will be widened to 6.5'.
- The northerly side slope for both the eastern and western section was revised to 4 to 1 and the southerly side was revised to 3 to 1. The revised side slopes will make it easier to cut and mow the grass for long-term maintenance.
- Two new catch basins will be added for the paved parallel parking spaces – one for the eastern section and one for the western section. All discharges to the eastern and western sections of the BMP will terminate at new dry stacked stone/boulder headwalls with rip-rap outlets. In addition, the outlet from the BMP will also have a dry stacked stone/boulder headwall. This will allow for improved long-term maintenance.
- There will be 3 access points from the northerly side of the BMP to the southerly side of the BMP as follows (see Sheet C7):
 -  Access Point 1 is 6' wide and is located at the interface between Stormwater BMP B and the eastern section of the existing Stormwater BMP.
 -  Access Point 2 is 8' wide and is located approximately in the middle of the existing Stormwater BMP.
 -  Access Point 3 is 5' wide and is located at the interface between Stormwater BMP C and the western section of the existing Stormwater BMP.

Each of these access points will provide access to the southerly side of the BMP for long-term maintenance as well as access to Stormwater BMP's B and C for long-term maintenance.

- The southerly side of the existing BMP will be revegetated with native trees and shrubs. The ground cover will be seeded with New England Wetland Plants Inc. Showy Wildflower Seed Mix.
- The base of the BMP will be seeded with New England Wetland Plants Inc. Wetland Seed Mix.
- The side slopes of the BMP will be seeded with New England Wetland Plants Inc. New England Conservation/Wildlife Mix.
- A new sloped granite curb will be installed along the edge of the parallel parking spaces to prevent future snow plowing from impacting the stormwater BMP.

Stormwater BMP C/Parking Area C – 12 Cherry Hill Drive (see Sheet 7)

As part of the construction of Parking Area C a new Stormwater BMP C is being proposed to mitigate the pre-and-post development peak flows and water quality. Currently stormwater from the rear of the parking lot flows to the existing drainage swale that discharges near wetland flags 24E and 23E (see Sheet C2). Stormwater BMP C will intercept this flow to provide for water quality treatment.

Five (5) soil cores were hand excavated along the top of the slope in the area for Parking Area C. The soil cores showed that there were no A or B soil layers. These soil layers have been removed from this area and replaced with a loamy fill (5" to 11"). This area also has soil piles, debris, concrete and trash and has revegetated with invasive species. The Estimated Seasonal High-Water Table (S.H.W.T.) was determined to be approximately 22".

As part of the construction of Parking Area C, Millipore is proposing to regrade this area and revegetate with native vegetation, trees and shrubs (see Sheet C11 Buffer Zone Restoration Plan). The ground cover will be seeded with New England Wetland Plants Inc. Showy Wildflower Seed Mix. This work will occur within the 35' No Disturb Area as documented on Sheet C3. This area can remain in its current condition. The proposal is to improve the Buffer Zone in this area by:

- Removing the invasive species. This will include a long-term Operation and Maintenance Plan to monitor this area and to remove any invasive species that may reemerge.
- Remove all trash, debris, soil piles and regrade for future revegetation.
- Revegetate this area with native trees and shrubs.

- Reseed with surface with New England Wetland Plants Inc. Showy Wildflower Seed Mix.

Existing Stormwater BMP Improvement – 17 Cherry Hill Drive (see Sheet C6)

Currently there is an existing stormwater BMP that stores and treats stormwater runoff from the existing parking lot that parallels the existing manufacturing facility. There is a single 12" RCP that discharges to the existing wetlands. A rip-rap weir overflow (approximated elevation 61.25) is located near the 12" RCP outlet.

As part of the construction of Parking Lot A the existing BMP will be increased in volume from 11,453 cubic feet to 13,576 cubic feet (2,123 cubic feet increase). The work on the existing BMP will also include:

- Removing the invasive species. This will include a long-term Operation and Maintenance Plan to monitor this area and to remove any invasive species that may reemerge.
- Introducing a 4' wide water quality stone surface treatment swale 95' long.
- Construct outlet headwall with discharge sedimentation forbay and vegetated velocity reducer to improve water quality and ease of long-term maintenance.
- Revegetate this area with native trees and shrubs.
- Reseed with surface with New England Wetland Plants Inc. Showy Wildflower Seed Mix.

The Plantings for all impacted areas will include the following.

Native trees shall be a minimum of 4 ft. in height and (4) different species shall be selected from the plant list below:

Common Name	Latin Name
Red Maple	Acer rubrum
Sugar Maple	Acer saccharum
American Sycamore	Platanus occidentalis
Black Birch	Betula lenta
Black Oak	Quercus velutina
Red Oak	Quercus rubra
White Oak	Quercus alba

Flowering Dogwood	Cornus florida
Basswood	Tilia americana

Native shrubs shall be a minimum of 2-3 ft. in height and (4) different species shall be selected from the plant list below:

Common Name	Latin Name
Lowbush Blueberry	Vaccinium angustifolium
Highbush Blueberry	Vaccinium corymbosum
Rosebay Rhododendron	Rhododendron maximum
Inkberry	Ilex glabra
Mountain Laurel	Kalmia latifolia
Clammy Azalea	Rhododendron viscosum
New Jersey Tea	Ceanothus americanus
Ninebark	Physocarpus opulifolius
Mapleleaf Viburnum	Viburnum acerifolium
Witch Hazel	Hamamelis virginiana

Any changes or deviations to the above tree and shrub lists shall be approved by the Conservation Agent or Conservation Commission.

Pre- and Post-Development Drainage

Methodology

The pre and post hydrology study was conducted using **HydroCAD a Stormwater Modeling System**. The Runoff Curve Numbers were selected from the tables listed within the **Soil Conservation Service Technical Release 55**. The terminology used by **HydroCAD** is summarized below.

1. **Subcatchment** - refers to a relatively homogenous area of land that drains into a single reach or pond.
2. **Reach**- refers to a uniform stream, Channel, or pipe that conveys water from one point to another reach or pond.

3. **Pond**- refers to a pond, swamp, dam, or other impoundment that fills with water from one or more sources and empties in a manner determined by a weir, culvert, ex-filtration or devise(s) at its outlet.
4. A **Link** may be used to:
 - Enter a Hydrograph generated outside of HydroCAD,
 - Interconnect several routing diagrams,
 - Scale a hydrograph,
 - Split a hydrograph into two components for independent routing,
 - Define a fixed or tidal tailwater elevation.”

Four storm events were analyzed as part of the pre-development drainage study. They are the 2, 10, 25, 50, and 100-year twenty-four-hour rainfall events. The twenty-four-hour rainfall precipitation was taken from the National Oceanic and Atmospheric Administration (NOAA) (see HydroCAD print out.)

Pre - Development Drainage:

Figure D1 defines the drainage subcatchment areas under existing conditions. The stormwater runoff from the existing site flows to a total of 2 Design Points. They are:

Design Point A (DPA) – Stormwater that flows the southerly property line at 17 Cherry Hill Drive from Parking Area A (Subcatchments EX1A).

Design Pont C (DPC) – Stormwater that flows to the existing wetland area (Wetland Flags 1E – 36E) from Parking Area B and C, Parallel Parking Area and existing parking lot (Subcatchments EX1B, EX2B and EX1C).

With the soil type, existing surface conditions, and hydrologic soil group, the runoff curve numbers can be determined. (See Existing Conditions HydroCAD Printout, Appendix A).

The 2, 10, 25, 50 and 100-year 24-hour storm events were run. Table 1 summarizes the peak flows to Design Points A and C for Pre – Development conditions in cubic feet per second.

Table 1: Pre-Development Peak Flows (cfs) to Design Points A and C

Storm Event	Design Point A	Design Point C Parking Lot B	Design Point C Parking Lot C
2-year storm event (3.24")	1.69 c.f.s.	0.86 c.f.s.	0.67 c.f.s.
10-year storm event (5.10")	3.41 c.f.s.	1.73 c.f.s.	2.04 c.f.s.
25-year storm event (6.27")	4.52 c.f.s.	2.30 c.f.s.	3.05 c.f.s.
50-year storm event (7.12")	5.33 c.f.s.	2.72 c.f.s.	3.82 c.f.s.
100-year storm event (8.06")	6.23 c.f.s.	3.18 c.f.s.	4.70 c.f.s.

Peak Flows – Cubic Feet Per Second (c.f.s.)

Post - Development Drainage:

Figure D2 defines the drainage subcatchment areas under proposed conditions. The stormwater runoff from the proposed site flows to the same 2 Design Points outline in the Pre Development Drainage above.

Three new Best Management Practice Ponds (BMP's) have been introduced to control and treat the stormwater runoff from the proposed development. They are:

- BMP A for Parking Area A
- BMP B for Parking Area B
- BMP C for Parking Area C

Under the Massachusetts Department of Environmental Protections Stormwater Regulations, one of the first things that is requested of the proponents of projects subject to the Stormwater Regulations is to consider the use of Environmentally Sensitive Site Design and Low Impact Development Techniques as outlined below.

Environmentally Sensitive Site Design and Low Impact Development Techniques

The Wetlands Regulations, 310 CMR 10.04, and the Water Quality Certification Regulations, 314 CMR 9.02, define environmentally sensitive site design to mean design that incorporates low impact development techniques to prevent the generation of stormwater and non-point source pollution by reducing impervious surfaces, disconnecting flow paths, treating stormwater at its source, maximizing open space, minimizing disturbance, protecting natural features and processes, and/or enhancing wildlife habitat. The Wetlands Regulations, 310 CMR 10.04, and the Water Quality Certification Regulations, 314 CMR 9.02, define low impact development (LID) techniques to mean innovative stormwater management systems that are modeled after natural hydrologic features. Low impact development techniques manage rainfall at the source using uniformly distributed decentralized micro-scale controls. Low impact development techniques use small cost-effective landscape features located at the lot level.

Proponents of projects subject to the Stormwater Management Standards must consider environmentally sensitive site design and low impact development techniques to manage stormwater. Proponents shall consider decentralized systems that involve the placement of a number of small treatment and infiltration devices located close to the various impervious surfaces that generate stormwater runoff in place of a centralized system comprised of closed pipes that direct all the drainage from the entire site into one large dry detention basin.

MassDEP has established an “LID Site Design Credit” to encourage developers to incorporate LID techniques in their project. In exchange for directing runoff from roads and driveways to vegetated open areas, preserving open space with a conservation restriction, or directing rooftop runoff to landscaped or undisturbed areas, MassDEP allows developers to reduce or eliminate the traditional BMPs used to treat and infiltrate stormwater.

Incorporating environmentally sensitive design that uses the land to filter and recharge the water back into the ground and that reduces the amount of paved areas is a critical first step in creating sustainable development.

The project implemented the request for “Environmentally Sensitive Site Design and Low Impact Development Techniques” in the following ways.

- As stated in the Environmentally Sensitive Design Practices “*Proponents shall consider decentralized systems that involve the placement of several small treatment and infiltration devices located close to the various impervious surfaces that generate stormwater runoff in place of a centralized system comprised of closed pipes that direct all the drainage from the entire site into one large dry detention basin*”.

The proposed project has created a total of 3 stormwater BMP’s that abut the areas that will be developed. By doing this the project intern will then:

- minimize stormwater runoff peak flows (cfs),
 - peak velocities (fs),
 - will minimize pollutant loads,
 - and by minimizing pollutant loads will allow each of the BMP’s to better treat the stormwater runoff.
-
- The BMP’s are designed to create “*decentralized systems that involve the placement of a number of small treatment and infiltration devices located close to the various impervious surfaces that generate stormwater runoff in place of a centralized system comprised of closed pipes that direct all the drainage from the entire site into one large dry detention basin*”. This will also allow the stormwater to be dispersed more evenly to the adjacent wetland areas.
 - In addition, stone/grassed Low Impact Design swales will capture the parking lot runoff for Parking Area A (existing parking lot) and Parking Area B and C. The grassed roadway swales will increase water quality by:
 - Increasing stormwater infiltration through the stone surface.

- Removing TSS as the stormwater flows through the stone surface.

The grass swales will be constructed with wide and flat side slopes that will:

- Promote a less concentrated stormwater runoff flow generated from the proposed roadway.
- This will reduce the stormwater velocity.
- The reduced stormwater velocity will allow for greater stormwater infiltration and water quality treatment.
- Will fit into the natural landscape areas.
- Allow for easier long term maintenance.
- Minimize winter snow plowing impacts.

Please Note:

On site soil testing confirmed that the exiting soils had a Hydrologic Soil Group of C/D with a High Seasonal High-Water Table. These types of soils are not suited for stormwater infiltration. The Massachusetts Department of Environmental Protection (MassDEP) recognizes that it may be difficult to infiltrate the required recharge volume on certain sites because of soil conditions. For sites comprised solely of C and D soils and bedrock at the land surface, proponents are required to infiltrate the required recharge volume only to the maximum extent practicable. The introduction of the stone/grassed Low Impact Design Swales will provide for some stormwater infiltration, but it will be minimal. In addition, the restoration of the buffer zone associated with the construction of Parking Lot C will also improve the natural surface stormwater infiltration.

The same 2, 10, 25, 50, and 100-year 24-hour storm events were run for the Post Development Conditions. Table 2 below summarizes the peak flows to Design Points A and C for the Post Development conditions in cubic feet per second.

Please Note:

- When engineers refer to the CN Value or Runoff Curve Number, they are describing a soil type (in our case 5 soil types as discussed above), existing surface conditions (grass, roof, pavement) and the soils Hydrologic Soil Group (in our case A and B). With this information the Natural Resources Conservation Service (NRCS) Runoff Curve Numbers can be determined (CN Value).

The NRCS CN value is related to soil type, soil infiltration capability, land use (grass, roof pavement), and the depth of the seasonal high-water table. Pavement has a CN value of 98 or 98% of the water runs off the surface when it is thoroughly wetted. Grass, in good condition (grass over 75% of the surface - which is less than your typical lawn), and a Hydrologic Group Classification of B has a CN value of 61 or 39% of the stormwater infiltrates when the soil is thoroughly wetted.

- Please note that any subcatchment that resulted in a Time of Concentration (T_c) of less than 5 minutes was entered as Direct Entry with a corresponding T_c of 5 minutes. For these subcatchments the Longest Flow Paths are not shown.

Table 2: Post-Development Peak Flows (cfs) to Design Points A and C

Storm Event	Design Point A	Design Point C Parking Lot B	Design Point C Parking Lot C
2-year storm event (3.24")	1.36 c.f.s.	0.80 c.f.s.	0.68 c.f.s.
10-year storm event (5.10")	2.16 c.f.s.	1.26 c.f.s.	0.97 c.f.s.
25-year storm event (6.27")	2.61 c.f.s.	1.53 c.f.s.	1.13 c.f.s.
50-year storm event (7.12")	2.89 c.f.s.	1.66 c.f.s.	1.24 c.f.s.
100-year storm event (8.06")	3.15 c.f.s.	1.79 c.f.s.	1.48 c.f.s.

As can be seen from Table 2 above there will be no increase in the pre and post development flows to Design Points A and C. A .01 cubic feet per second increase does occur for the 2-year storm event for Design Point C for Parking Lot C. But only an engineer would consider this an increase.

The drainage system (pipes) has been designed to convey the 100-year storm event.

If you have any questions or concerns, please feel free to contact me at 978-500-8419

Sincerely,



ASB design group, LLC
Thad D. Berry, P.E.
Principal

